06. S-matrix bootstrap with an explicit gravitational infrared prescription

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Suppose we do not know a theory's microscopic machinery, but do know that collision probabilities must be sensible and different descriptions of the same process must agree. Can those rules alone rule out a proposed interaction strength? This problem searches the space of scattering possibilities rather than choosing one detailed microscopic theory. Because gravity acts over long distances and allows very low-energy radiation, defining the scattering question correctly is itself part of the work.

Key terms

- **S-matrix** — The object relating incoming quantum states to outgoing states in a scattering experiment.
- **Partial wave** — A contribution to scattering organized by angular momentum, which makes some probability constraints easier to state.
- **Unitarity** — Conservation of total probability; an allowed scattering process cannot produce a total probability greater than one.
- **Elastic scattering** — Scattering that stays within the chosen particle channel without producing additional particles or excitations.
- **Crossing symmetry** — Relations between amplitudes for processes with differently assigned incoming and outgoing particles.
- **Infrared prescription** — A stated way to handle low-energy radiation and long-distance singularities.
- **Outer relaxation** — A larger, easier-to-test set containing all valid possibilities under the assumptions. Excluding a point from it can exclude the original possibility.
- **Inner ansatz** — A selected subset of candidate amplitudes; failing to find a solution there need not exclude other amplitudes.
- **Dual certificate** — An independently checkable mathematical witness establishing a bound or inconsistency.

Why this matters

The method could identify restrictions shared by many possible microscopic theories. That is useful when no single theory is established. Its reliability depends on not mistaking a limitation of the search formula, grid or cutoff for a limitation imposed by physics.

What progress would mean

A certified exclusion would eliminate a parameter choice under the stated scattering assumptions. The gravity-free reference case first checks the machinery; adding a defensible treatment of gravity would be the important next step.

Core question

Can one certify an exclusion for a low-energy scalar-scattering parameter using a crossing-symmetric partial-wave relaxation whose infrared and truncation assumptions are explicit?

Scope and assumptions

Begin with identical massive scalars in four dimensions and no gravity to validate the machinery. The gravitational stage must first define an infrared-finite observable or a justified regulator/pole treatment, including massless radiation. Keep one scalar mass and one amplitude coefficient fixed; optimize a second coefficient in a declared normalization.

This is complementary to Challenge 02: the focus is amplitude feasibility and partial-wave constraints rather than deriving a low-energy positivity sum rule. A nongravitational baseline does not count as solving the gravitational stage.

Mathematical target

Choose $A = 16\pi \sum_{\ell\geq 0}(2\ell+1)a_\ell(s)P_\ell(\cos\theta)$, s+t+u=4m^2, $\rho=\sqrt{}(1-4\mathrm{m}^2/\mathrm{s})$, and S_$\ell$=1+2i$\rho$a_$\ell$. Then |S_$\ell$|$\leq$1 is equivalent to Im a_$\ell\geq\rho$|a_$\ell$|^2. Equality needs an actually elastic sector.

State how crossing is imposed and distinguish an outer relaxation of necessary constraints from an inner finite amplitude ansatz. A dual exclusion from an outer relaxation transfers to the full problem only when every physical amplitude maps into that relaxation. Bound energy, angular-momentum, and basis tails; checking a finite mesh is insufficient.

Required outputs and proof obligations

Export the amplitude conventions, IR prescription, finite optimization problem, mapping from the continuum problem, and dual certificate with its positivity/error margins. Feasible ansatz points are examples satisfying the verified constraints only. If the IR construction or tail control fails, state which conclusion remains unavailable.

Validation and rejection controls

Check free scattering and a specified analytic nongravitational benchmark. Verify the unitarity disk algebra independently. Test between grid points and beyond cutoffs using proved bounds. Demonstrate that known admissible examples are not falsely excluded. Increasing precision is a diagnostic; rigorous enclosures are needed for certification.

Milestones and research extension

Baseline: certified finite nongravitational relaxation. Strong: justify an IR treatment and certify one conditional gravitational exclusion. Research extension: improved continuum bounds or a constructive amplitude with a precisely delimited consistency claim.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Alberte et al., Positivity Bounds and the Massless Spin-2 Pole — Infrared and analyticity issues that must be resolved before importing standard bounds.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$A = 16\pi \sum_{\ell \geq 0} (2\ell + 1) a_\ell(s) P_\ell(\cos\theta), \quad \rho = \sqrt{1 - 4m^2/s}, \quad S_\ell = 1 + 2i\rho a_\ell.$$

$$|S_\ell| \leq 1 \quad \Longleftrightarrow \quad \operatorname{Im} a_\ell \geq \rho |a_\ell|^2 \qquad (s > 4m^2).$$

Equality requires an elastic sector. A certified outer relaxation can exclude a physical amplitude only after the inclusion of every admissible amplitude and all tail bounds are proved. Failure of an inner ansatz excludes only that ansatz. The gravitational stage has a separate infrared prerequisite.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — elastic window with gravity (68–79).** Massless radiation removes the simple massive-theory inelastic threshold assumption. The equality form of elastic unitarity cannot be silently reused. The relaxed inequality and the IR observable must be justified.
2. **Blocking — singular partial waves.** Long-range graviton exchange makes naive four-dimensional angular projection problematic. The proposed all-purpose Roy-like equation (89–92) is not a derivation of the required kernels.

3. **Major — inner versus outer approximation.** Infeasibility of a restricted amplitude ansatz excludes only that ansatz. This distinction is essential for every claimed no-go theorem.
4. **Major — missing continuum coverage.** A finite partial-wave/energy grid cannot establish all-energy unitarity. The new specification requires analytic tail bounds and otherwise reports a finite relaxation, preserving meaningful progress without overstating it.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **Can one certify an exclusion for a low-energy scalar-scattering parameter using a crossing-symmetric partial-wave relaxation whose infrared and truncation assumptions are explicit?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: certified finite nongravitational relaxation. Strong: justify an IR treatment and certify one conditional gravitational exclusion. Research extension: improved continuum bounds or a constructive amplitude with a precisely delimited consistency claim.

Sources supporting the corrections

- Alberte et al., Positivity Bounds and the Massless Spin-2 Pole — Infrared and analyticity issues that must be resolved before importing standard bounds.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.

- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.